

 Yildiz Technical University Faculty of Chemical and Metallurgical Department of Mathematical Engineering Final Exam Question and Solution Sheet	SCORE TABLE						
	1	2	3	4	5	SUM	
Name-Surname							
Number							
Department					Date	05.06.2023	
Lecture	ALGEBRA	Grup Number	2	Duration	90 dak.	Room	Online
Lecturer	Assist. Prof. Dr. Serkan ONAR				Signature		
<i>You must show all details of your work to get full credit. Correct answers with no detail will get zero credit.</i>							

1) A. Write the elements of dihedral group D_4 with respect x and y . Is D_4 a p-group? Justify your answer.

1) B. Determine or whether $(C_6, \cdot)$ be a cyclic group with order 6 is a normal subgroup of dihedral group D_6 .

Find the elements of quotient group D_6 / C_6 .

2) A. Show that group $(\mathbb{Z}_8^*, \cdot)$ is a isomorphic to $V_4 = Klein - 4$ group. (By examining the elements and orders)

2) B. Let G be a commutative group and $n \in \mathbb{Z}^+$. Prove or disprove that $H = \{a \in G \mid x \in G \text{ için } a = x^n\}$ is a subgroup of G ? Explain your answers.

3) A. Define the notion of subring and ideal. Every ideal is a subring? Is the converse true? Explain.

3) B. Let $I_2(\mathbb{Z}) = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{Z} \right\}$ be set of all upper triangular matrices.

Prove or disprove that $I = \left\{ \begin{bmatrix} 0 & b \\ 0 & c \end{bmatrix} \mid b, c \in \mathbb{Z} \right\}$ is an ideal of $I_2(\mathbb{Z})$.

4) Let $\mathbb{Z}_{15}$

- Find Sylow p-subgroups.
- Determine $\mathbb{Z}_{15}$ as an internal direct product of Sylow p-subgroups.
- Write $\mathbb{Z}_{15}$ as an external direct product.
- Find all automorphisms of $\mathbb{Z}_{15}$.

5) A. Define the dihedral group D_5 and write the elements. Find the order of element yx^3 and compute commutator $[x, yx^3] = ?$

5) B. Find the elements (a, b) with order 6 in direct product group $\mathbb{Z}_4 \times \mathbb{Z}_6$. Determine the number of elements.

(Hint : $lcm(o(g_1), o(g_2)) = 6$)